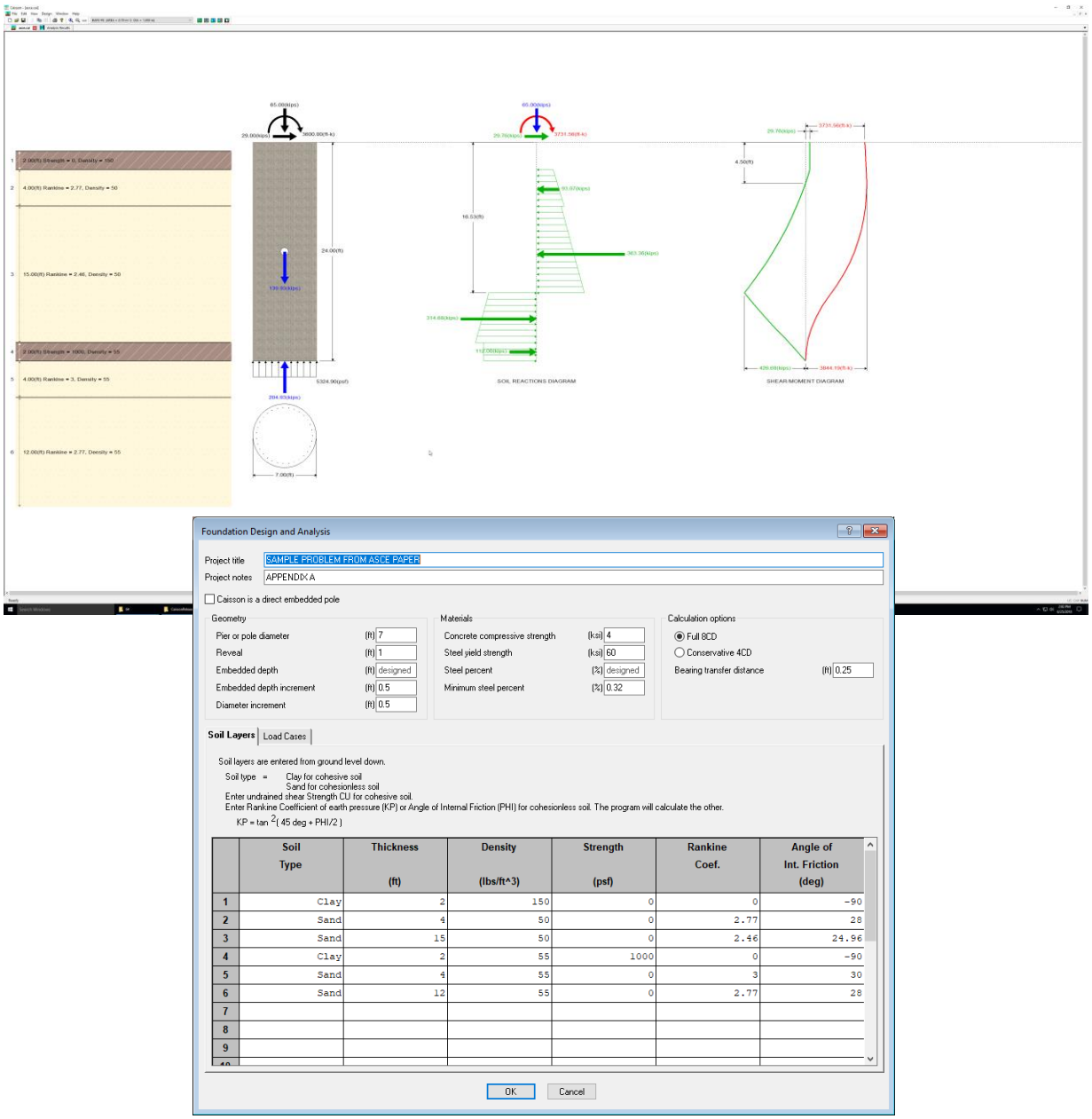
CAISSON



	Soil Type	Thickness (ft)	Density (lbs/ft^3)	Strength (psf)	Rankine Coef.	Angle of Int. Friction (deg)
1	Clay	2	150	0	0	-90
2	Sand	4	50	0	2.77	28
3	Sand	15	50	0	2.46	24.96
4	Clay	2	55	1000	0	-90
5	Sand	4	55	0	3	30
6	Sand	12	55	0	2.77	28
7						
8						
9						

POWER LINE® SYSTEMS

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(Printing date: April, 2021)

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1. OVERVIEW

CAISSON is a Microsoft Windows program for the design of moment resisting concrete pier foundations for pole structures, using the ultimate strength design concept. It can also be used for the design of direct embedded poles. In such cases the diameter of the pier is that of the pole and the concrete/reinforcing properties are irrelevant. The theoretical basis of the program is described in the paper of Appendix B. There is no calculation of deflection under service loads. **CAISSON** is a moment resisting analysis program and does not verify the vertical uplift or compression capacity of the foundation or of direct embedded poles. Since a majority of transmission line structures foundations are controlled by lateral loads, this is generally acceptable. The end bearing pressure is given in the report and compared to compression capacity. When the pole is part of an H-Frame structure, uplift is a major factor and must be checked independently of this program as well.

It cannot be overemphasized that **CAISSON** is to be used by the Foundation Engineer only. He/she should possess a basic understanding of soil mechanics, how to read and interpret soil boring logs and other soils data provided by the geotechnical engineer. Soils data should include the weights of dry or submerged soils (if any) at various depths, the coefficients of internal friction for non-cohesive soils, and the value of cohesion for cohesive soils.

The output of **CAISSON** provides echo checks, soils resistance diagrams, shear and moment diagrams, the required length of the pier and its steel reinforcement. The program proposes arrangements for different bar sizes in US or SI units.

CAISSON is fast and interactive. Therefore, the effects of changes in pier diameter, safety factor and material strengths can be studied at a low cost.

2. DISTRIBUTION, MAINTENANCE, AND TECHNICAL SUPPORT

CAISSON is developed, maintained and supported by:

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E-mail: support@powline.com
Web site: https://www.powerlinesystems.com

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To receive information regarding updates and new versions you should sign-up for our newsletter. Sign-up instructions are included on our web site.

3. HARDWARE REQUIREMENTS AND LIMITATIONS

To run **CAISSON** you need Microsoft Windows 7, 8, 8.1, 10 or newer (x64 edition required).

4. INSTALLATIONS

CAISSON is shipped exclusively by e-mail. The program and files can be installed in any directory.

For E-Mail installation:

Step 1) Use your web browser to save the e-mailed URL to a temporary directory on your computer. The download size will be approximately 4 MB.

Step 2) Unzip the cdsetup.zip file (copy the resulting setup folder to desired drive for archive and re-install purposes).

Step 3) Run the resulting setup.exe program

Step 4) Follow the instructions provided by Setup

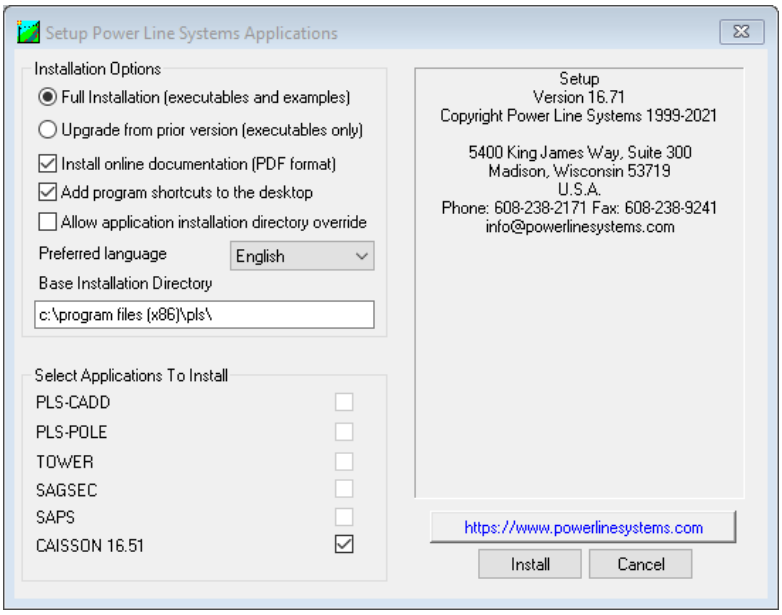


Fig. 4.1 Setup Power Line Systems Applications

After completing the above four steps you should see the **CAISSON** icon appear in the **PLS-CADD** group. You can run **CAISSON** by clicking on its icon in the **PLS-CADD** group or by selecting its icon with **Start/ Programs/ CAISSON**.

4.1 Subscription Based Licensing

In version 16.00, Power Line Systems introduced a fundamentally new way of licensing its software which is referred to as Subscription Based Licensing (SBL). SBL replaces the hardware key-based licensing that has been in effect since 1993 and the USB hardware key is no longer available or supported by Power Line Systems.

SBL is a named user subscription license that entitles you to run PLS software for some period which is generally one year. The SBL will not function after the expiration date and the subscription for the software must be renewed to continue to use the PLS software. Starting 60 days from the end of your subscription period the program will warn you that it will cease functioning at the end of 60 days. These warnings become more frequent at 30 days. When zero days are left the software will cease functioning. Power Line Systems will also send an email to you 30 days from expiration.

PLS will provision licenses for you on the PLS Subscription. You will be provided with a link to download your software as a ZIP file. This ZIP file contains the information necessary to view and claim licenses for your organization from our servers. Alternately, you may be emailed a ".plsreg" file that contains this information. In the latter case, your PLS software will prompt you for the ".plsreg" file and upon receiving it will show you all licenses available to your organization and allows you to claim one. When claiming a license, users are required to provide your name, phone number and email address. End users need to claim with their own email address and contact information. Generic User names and email addresses are not permitted.

The license will be reserved for your use until you request that it be released. Licenses may be transferred between employees of the same organization once every 30 days. The PLS software is controlled by a single SBL so a user does not typically need to claim multiple licenses.

4.1.1 Internet Requirements for the SBL

SBL entitlements are provisioned by Power Line Systems and served exclusively via the PLS license server. Third party license servers are not supported. The PLS software must be able to periodically communicate with our license server via port 443 at https://balancer.plscloud.com (not a human accessible web site) in order to operate.

The Subscription Based License is required to check into our license server once every seven days. Each time the software connects it will automatically renew its license for the next seven days. The software can run without an internet connection if there has been a connection with the license server within the last 7 days. If this period passes, then a user will need reestablish an internet connection and allow the PLS software to check in with our license server to continue using the SBL. If there are any questions on the SBL please feel free to email the PLS Technical Support team at support@powerlinesystems.com.

Virtualized or Server (Remote Desktop Services/Citrix) use must always have an Internet connection to function. Since a connection is also necessary to reach the server hosting the software this is not expected to be an obstacle to adoption. Virtual Machines such as Oracle VirtualBox, Parallels Desktop, VMWare, Microsoft Hyper-V and others require an Internet connection for SBL use.

4.1.2 The SBL license file

The Power Line Systems Subscription Based License (SBL) can be identified by the .plsreg file extension. A user will have to claim a SBL with a valid subscription to run the software. The name of the license file identifies the number and capabilities of each license type with the number and letter designation at the end of the file name.

For example 62170-SBL-2-V.plreg is a file that represents 2 copies of PLS-CADD Survey with the PLS client ID of 62170. As another example the 62170-SBL-3-ISW.plsreg file represents 3 copies of PLS-CADD/Lite with PLS-POLE/Steel and PLS-POLE/Wood.

The letter designations for each program within the license file are as follows:
(P) PLS-CADD
(O) Optimum Spotting
(V) PLS-CADD/Survey
(I) PLS-CADD/Lite
(A) SAPS (Finite Element Sag Tension)
(T) TOWER
(R) EDF Option for TOWER
(S) PLS-POLE/Steel
(W) PLS-POLE/Wood
(C) PLS-POLE/Concrete
(L) PLS-POLE/Laminated Wood + Mast
(F) PLS-POLE/FRP
(E) CAISSON
(D) PLS-CADD/Distribution
(G) SAGSEC
(X) PLS-GRID Server
(Z) PLS-GRID Desktop Client

4.1.3 Claiming and Unclaiming a SBL

When you first receive a full installation of the PLS software there are installation instructions in the form on an email which contains a download link for the installation files for the software. The software can be installed as described in section 4.0 of this manual. The download link will include the PLS installer and license files for the different license options within your organization.

The SBL license model is not a floating or network license that can be shared. It is referred to as a named single user license. It is tied to an end user's email credentials as well as their computer. A single user can install this license file on up to two computers with the idea being a

user can work on a computer at their office or a computer at their home or in the field. The SBL is also supported on virtual machines.

The link, https://www.powline.com/internal/Claim_License_PLS.pdf, describes each step for installing the PLS software and claiming the Subscription Based License. After a successful installation the PLS software you can then launch any of the programs which will bring up the PLS license dialog, see Figure 4.2. A user can always get to the License dialog by using the small key button next to the red X in the About Dialog and it will bring you back to the License dialog.

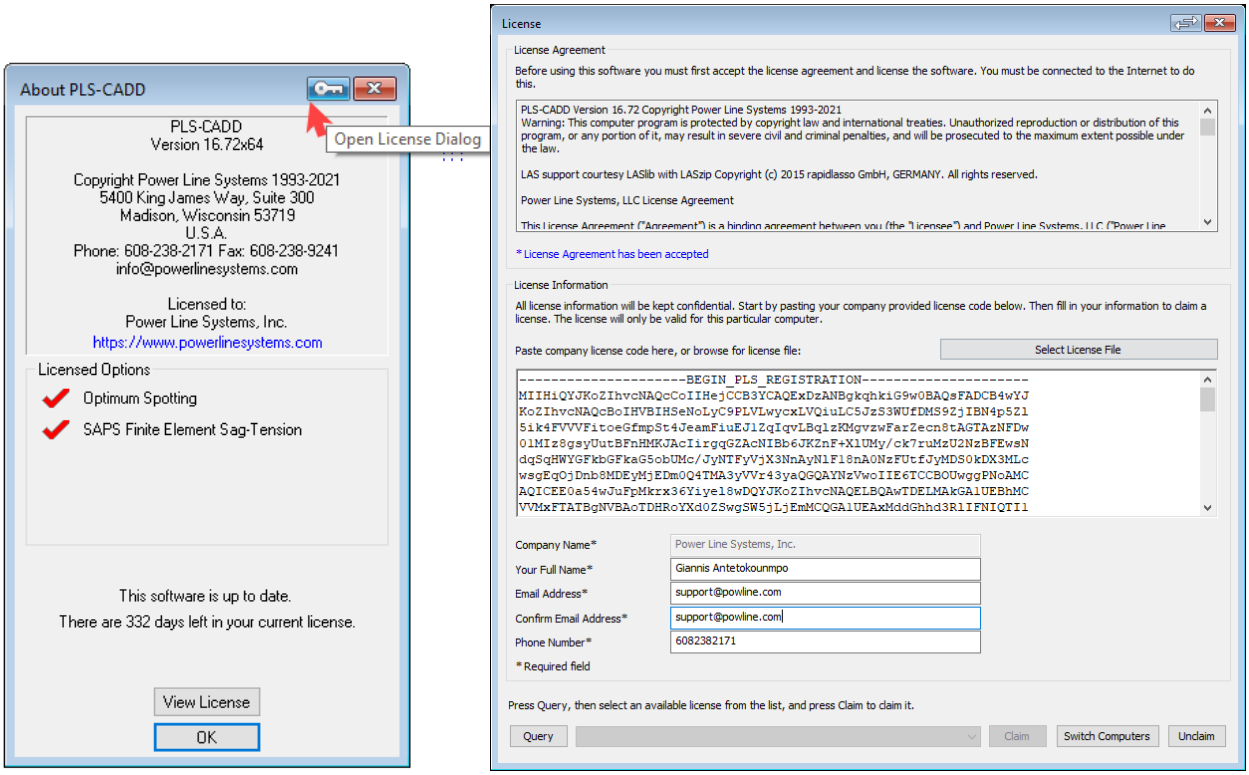


Fig. 4.2 PLS License Dialog

The first step in claiming a license is to read and accept the license agreement at the top of the dialog. For the accept button to become active you must read the entire license agreement and scroll to the bottom. The next section is the license information. You will need to select the appropriate license file that matches the software capabilities required. If the license you wish to claim is not in the pull down list and you wish to claim another license set you will need to use the Select License File button and select the alternative .plsreg file In order to claim one of these licenses. The end user also is required to enter your Full Name, Email Address, and Phone Number which enables the Query button. After selecting Query the list of available licenses that have not been claimed will be shown and you can select one of the licenses to claim. This will then launch the program and the About dialog will appear and which you can OK to continue to the program.

The *Switch Computers* button will allow a user to switch the SBL between the primary and secondary machine. So if a user leaves the office and wants to work from home, they will need to use this Switch Computers button on their home machine to toggle the license there.

The *Unclaim* button will allow a user to release their license. This will clear out the user information from our license server about this machine and user for this SBL, and allow this license to be available for others within the organization to claim. This button can also be used in the case where a user has received a new machine. A user cannot use the *Unclaim* button if they have claimed the license within the last 30 days.

4.2 Electronic Manual and Online Help

If you have installed the electronic version of the manual (PDF format) by checking *Install online documentation* in Fig. 4.1, you will have access to online help in any dialog box. All you need to do is to click on the "?" button at the top right of the box and you will be taken to the relevant section of the electronic version of the manual where you will find the appropriate information. Updates to the manual are periodically made available via our Internet site. You should use the **Help/Check for Updated Manual** command to discover if they are available and download them. They are also advertised in our Newsletter which you can read and subscribe to on our web site.

A Quick Search Toolbar is available when the **View/ Toolbars/ Quick Search Toolbar** is selected. You may enter text in the quick search toolbar which will be used to search all menu items in the program, sections of the manual, online technotes, File Preferences Settings, and online videos. If you click on any of these items then it will take you to that particular selection whether it is a menu command, section in the manual, video, preference setting, or technote. Figure 4.3 shows the resulting menu items and help topics in the manual with the word “post” as the search string. You can limit your search to certain sections by prefixing your search term with ‘m’ for menu item, ‘h’ for help topics in the manual, ‘t’ for technotes, and ‘v’ for videos. The search dialog can also be used as a calculator by prefixing your calculation with the ‘=’ sign.

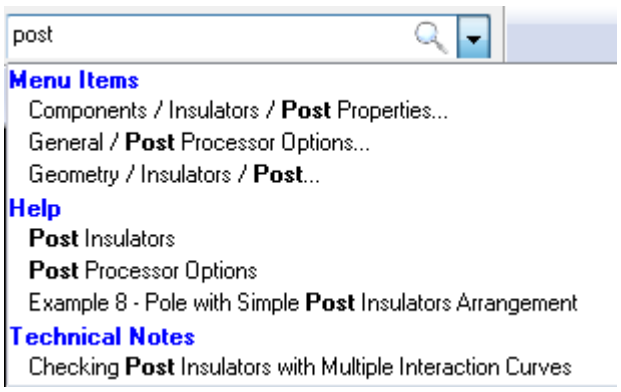


Fig. 4.3 Quick Search Toolbar

You can enter ‘=help’ to display a list of the available arithmetic operations and functions available. Some of these functions include the ability to calculate the moment of inertia and section modulus of rectangular, round, and pipe cross sections which are useful when populating these inputs in the crossarm, brace, davit arm, etc. component tables.

For more on the use of the Quick Search Toolbar you can see the video at http://www.youtube.com/watch?v=DwYgf5e9lOk&feature=youtu.be

5. GETTING STARTED

This section will show you how to load an existing pier model, briefly look at the various editing options, and view the results. You should be familiar with the basic Windows commands and terminology. Start **CAISSON** by clicking on the CAISSON icon.

5.1 Loading and Saving Files

Loading and saving data for a pier model can be done by clicking on the **File/ Open** or **File/ Save** menus. With **File/ Open**, the dialog of Figure 5.1 will appear. You can specify a file name by typing it in the filename field or by clicking on a file name in the list and then clicking on **OK**. You can switch to another directory by double clicking on its name in the directory list. The directory that appears when you first get to the dialog of Figure 5.1 can be specified in the preferences dialog described in Section 5.2. For example, open the file **asce.cai** (full example included in Appendix A). All **CAISSON** input files end with the **.cai** extension.

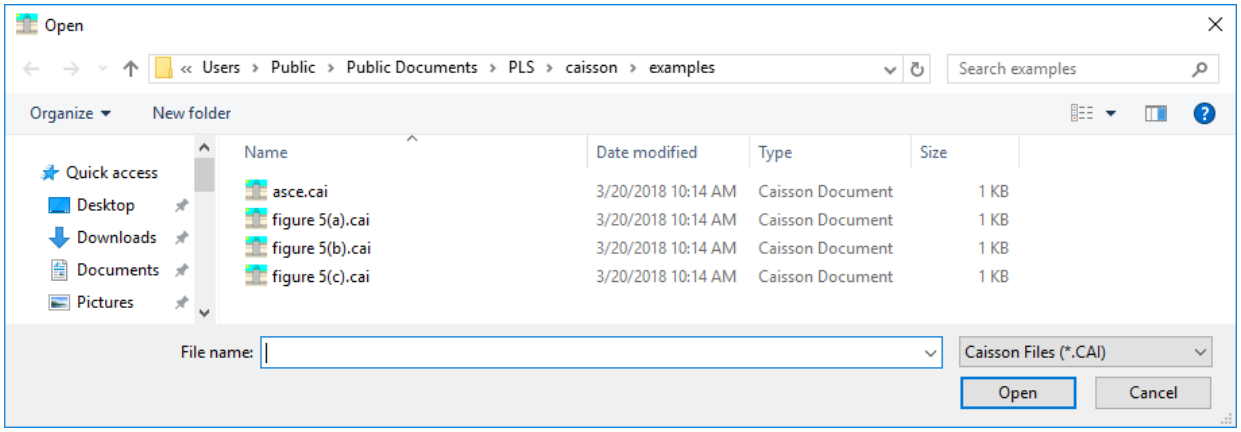


Figure 5.1 Opening Files

5.2 Directory Preferences and Units

The dialog of Figure 5.2 is accessed with **File/ Preferences**. It is used to select the default directories for the Application (the executable program) and the Projects (the pier models). This dialog is also used to select the units system and other settings for the use of the program such as the report font or graphics font.

CAISSON can accept input and generate output in both SI (metric) and US customary units. A model created in US units can be viewed in SI units and vice versa.

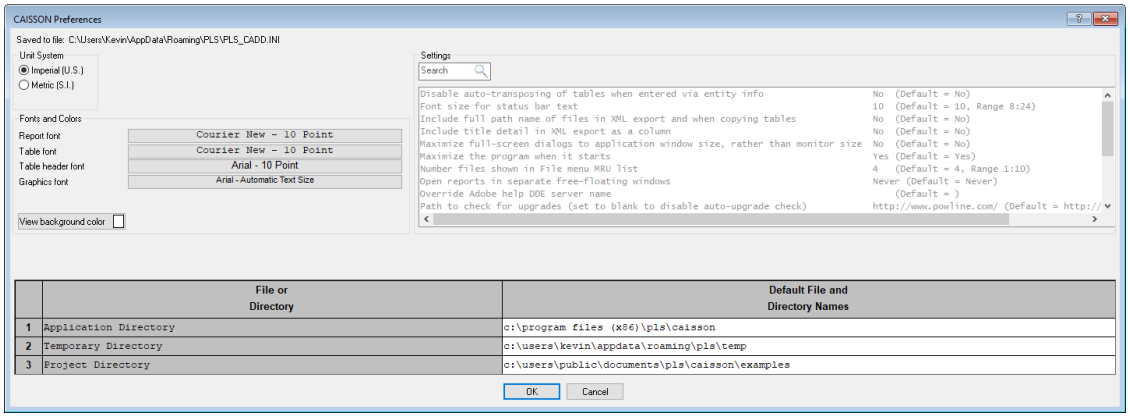


Figure 5.2 Preferences Dialog

5.3 Creating and Editing a Pier Model

Now that you have selected the file **asce.cai** you will see a figure of the soil profile depicted in Figure 5.3 when the file is loaded.

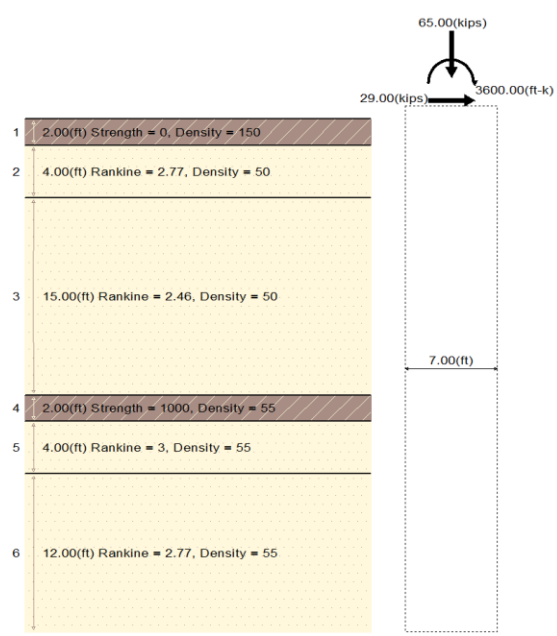
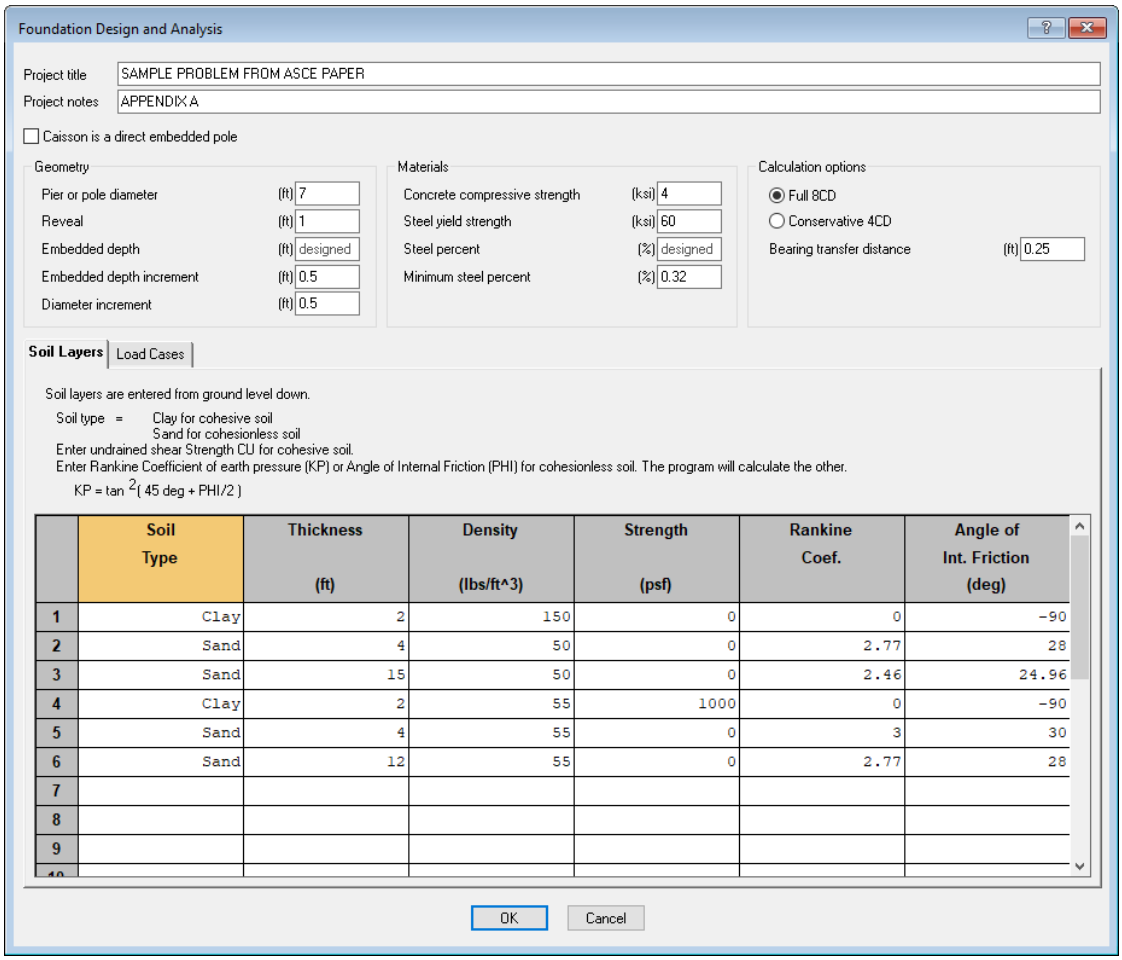


Fig. 5.3 Soil Profile

You can view or edit its content by using the **Design/Foundation Design and Analysis** command. Figure 5.3.a shows the Foundation Design and Analysis dialog which contains foundation, soil and load cases parameters. The following section describes the design inputs, section 6.0 describes the soil data and section 7.0 describes the design load data.



	Soil Type	Thickness (ft)	Density (lbs/ft^3)	Strength (psf)	Rankine Coef.	Angle of Int. Friction (deg)
1	Clay	2	150	0	0	-90
2	Sand	4	50	0	2.77	28
3	Sand	15	50	0	2.46	24.96
4	Clay	2	55	1000	0	-90
5	Sand	4	55	0	3	30
6	Sand	12	55	0	2.77	28
7						
8						
9						

Fig. 5.3.a Foundation Design and Analysis Dialog

Project Title: Project title to display within reports.

Project Notes: Project notes to display within reports.

Caisson is a direct embedded pole: If not selected calculations will assume the foundation is a concrete pier. If selected the calculations will assume the foundation is a direct embedded pole.

<u>Geometry</u>

Pier or pole diameter: Pier diameter or pole diameter (for direct embedded poles). If *designed* in shown or zero inputted then CAISSON will calculate this value based on other inputs.

Reveal: Pier reveal above the ground.

Embedded depth:	Embedded depth of foundation. If *designed* in shown or zero inputted then CAISSON will calculate this value based on other inputs.
Embedded depth increment:	Increment along the foundation to check and report results.
Diameter increment:	Diameter of the foundation to check or report results.

<u>Materials</u>

Concrete compressibe strength:	Self explanatory
Steel yield strength:	Self explanatory
Steel percent:	Specified percent of steel needed for pier. If *designed* in shown or zero inputted then CAISSON will calculate this value based on other inputs. See section 8 of CAISSON manual.
Minimum steel percent:	Minimum specified percentage of steel used.

<u>Calculation options</u>

Full 8CD:	See section 7 of CAISSON manual.
Conservative 4CD:	See section 7 of CAISSON manual.
Bearing transfer distance:	If bottom of foundation is within the bearing transfer distance then the bearing capacity will be calculated using the bottom layer.

Soil Layers tab:	See section 6 of CAISSON manual.
Load Cases tab:	See section 7 of CAISSON manual.

5.4 Executing Analysis and Viewing Results

Once the data is prepared, the design is performed with the command **Design/Run Analysis**. After the design is completed you will be taken to the Analysis Results report window where you can view the entire output file and also a figure showing the calculated reactions for the pier shown in Figure 5.4. Right clicking with the mouse inside the report window will present a menu with several report options described below:

Save as lets you send the output to any file.
Append to lets you append the output to the end of an existing file.
Font lets you change the fonts for viewing the output window.
Autosize Font will pick the largest font for which none of the lines wrap onto another line.
Print sends the report to the printer.
XML Export allows you to export a xml file of the results.
Table View allows you to view the results in a tabular format

You will also be able to see calculated reactions for the pier as shown in Figure 5.5.

Analysis Results

```
*****************************************************************************************
*                                                                                       *
* CAISSON - Pier Foundations Analysis and Design -  Copyright Power Line Systems, Inc. 1993-2018 *
*                                                                                       *
*****************************************************************************************

Project Title : SAMPLE PROBLEM FROM ASCE PAPER
Project Notes : APPENDIX A
Project File  : ...\CAISSON\ASCE.CAI
Date run      : ... July 2018
by            : CAISSON Version 15.30
Licensed to   : Power Line Systems, Inc.

-----------------------------------------------------------------------------------------
Input Data
-----------------------------------------------------------------------------------------

Calculation Options
--------------------
Calculation method    : Full SCD
Bearing transfer      :     0.25 (ft)
Minimum steel percent :     0.32 (%)

Pier Properties
---------------
Diameter              :     7.00 (ft)
Embedded depth        :     N.A. (calculated - see load case results)
Steel area percent    :     N.A. (calculated - see load case results)
Reveal                :     1.00 (ft)
Concrete strength     :     4.00 (ksi)
Steel yield           :    60.00 (ksi)
```

Soil Properties

Layer	Type	Thickness (ft)	Depth at Top of Layer (ft)	Density (lbs/ft^3)	CU (psf)	KP	PHI (deg)
1	Clay	2.00	0.00	150.0			
2	Sand	4.00	2.00	50.0		2.770	28.00
3	Sand	15.00	6.00	50.0		2.460	24.96
4	Clay	2.00	21.00	55.0	1000.0		
5	Sand	4.00	23.00	55.0		3.000	30.00
6	Sand	12.00	27.00	55.0		2.770	28.00

Loads at Top of Pier

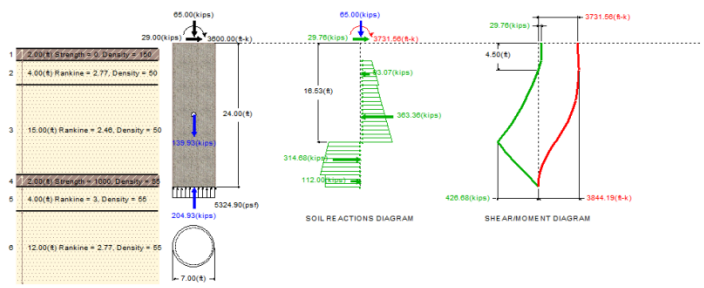
Load Case	Moment (ft-k)	Axial Load (kips)	Shear Load (kips)	Soil Strength Factor
	3600.0	65.0	29.00	1.000

```
-----------------------------------------------------------------------------------------
Analysis Results
-----------------------------------------------------------------------------------------
```

Load Case Results

Load Case	Diameter (ft)	Embedment Depth (ft)	Length (ft)	Minimum Steel Percent (%)	Minimum Steel Area (in^2)	Weight (kips)	Moment Capacity (ft-k)	Axial Capacity (kips)	Axial Load Pressure (ksf)	Weight Pressure (ksf)	End Bearing Pressure (ksf)	Maximum Pier Moment (ft-k)	Maximum Pier Shear (kips)	Total Soil Moment (ft-k)	Total Soil Shear (kips)	Soil Bearing Strength (ksf)	Status
	7.00	23.000	24.000	0.44	24.38	139.926	3922.2	65.5	1.69	3.64	5.32	3844.2	426.7	3731.56	29.76	12.24	OK

The controlling load case is ""



Ultimate Resisting Forces Along Pier for Load Case ""

Type	Distance of Top of Layer to Top of Pier (ft)	Thickness (ft)	Density (lbs/ft^3)	CU (psf)	KP	Force (kips)	Arm (ft)
Clay	1.00	2.00	150.0			0.00	2.00
Sand	3.00	4.00	50.0		2.770	93.07	5.17
Sand	7.00	9.53	50.0		2.460	363.36	12.28
Sand	16.53	5.47	50.0		2.460	-314.68	19.38
Clay	22.00	2.00	55.0	1000.0		-112.00	23.00

Shear and Moments Along Pier for Load Case ""

Distance below Top of Pier (ft)	Shear (kips)	Moment (ft-k)
0.00	29.8	3731.6
1.00	29.8	3761.3
1.20	29.8	3767.3
2.40	29.8	3803.0
3.00	29.8	3820.8
3.60	18.8	3835.4
4.50	0.0	3844.2
4.80	-6.4	3843.3
6.00	-35.7	3818.5
7.20	-68.5	3756.0
8.40	-102.0	3654.1
9.60	-139.2	3509.7
10.80	-180.1	3318.5
12.00	-224.8	3075.9
13.20	-273.1	2777.6
14.40	-325.2	2419.0
15.60	-381.0	1995.7
16.53	-426.7	1621.1
16.80	-412.9	1506.9
18.00	-349.6	1049.0
19.20	-282.7	669.3
20.40	-212.0	372.1
21.60	-137.6	161.9
22.80	-67.2	40.3
24.00	0.0	0.0

Fig. 5.4 Analysis Results Report Window

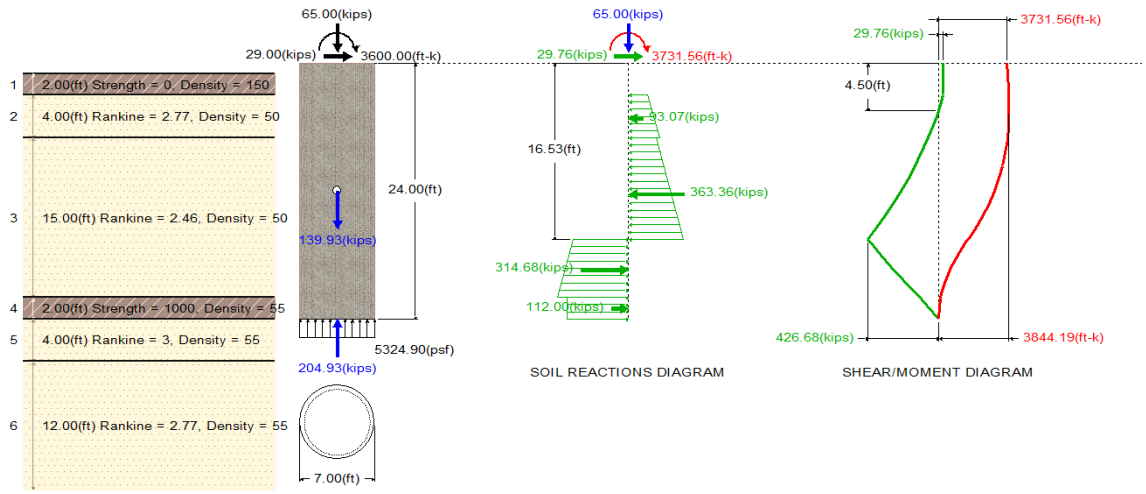


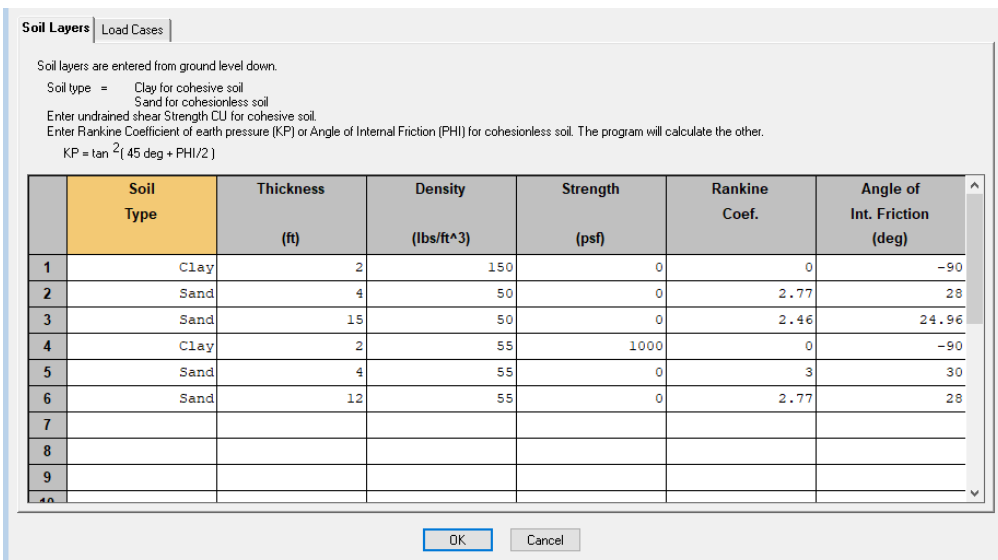
Figure 5.5 Calculated Reactions

5.4.1 Exporting DXF

A dxf drawing of the results as shown in Figure 5.5 can be created by going to **File/Export DXF** where the program prompts for a file name and location to save the file.

6. SOIL DATA

The design procedure requires that soil properties for various layers at the site of the proposed foundation be input (Figure 6.1). These properties are obtained by a standard soil boring program. Two typical soils are considered. Cohesive soils that are characterized by their cohesive strength C_U will be denoted soils of type Clay. Non-cohesive soils that are characterized by their angle of internal friction phi are denoted type Sand. The soil pressure at any depth can be obtained from the density and depth of the layers above. As shown in the paper of Appendix B, the caisson is assumed to be a rigid body. The pier is assumed to have failed when any additional load cannot be resisted by the surrounding soil. Assumptions for determining ultimate soil reactions (see Fig. 5 in Appendix B) are based on the soil cohesion C_U for cohesive soils, and on the soil pressure P_i and its angle of internal friction phi, or equivalently its Rankine coefficient K_p, for non-cohesive soils.



	Soil Type	Thickness (ft)	Density (lbs/ft^3)	Strength (psf)	Rankine Coef.	Angle of Int. Friction (deg)
1	Clay	2	150	0	0	-90
2	Sand	4	50	0	2.77	28
3	Sand	15	50	0	2.46	24.96
4	Clay	2	55	1000	0	-90
5	Sand	4	55	0	3	30
6	Sand	12	55	0	2.77	28
7						
8						
9						

Figure 6.1 Soil Properties

The authors of the paper who developed the original CAISSON program modified the method for calculating the soil reactions of a cohesive soil as stated in the Layered Theory section of Appendix B (and shown in Figure 5) because they felt the results were too conservative. Instead of the 4CD calculation for the soil reaction shown for cohesive soils that occur before a depth of 1.5D up to a depth of 3D and 8CD after the depth of 3D, they used 8CD to calculate the soil reactions of a cohesive soil independent of the layer depth. Both methods are available in **CAISSON** as of version 10.30. Either the Full 8CD method of calculation which uses 8CD for the calculation of cohesive soil reactions independent of depth or the Conservative 4CD method referred to in the paper may be selected in the **Design/Foundation Design and Analysis** dialog box shown in Figure 6.2 below. When starting a new project, **CAISSON** will default to the Full 8CD method, which has been in use in the program for over 20 years. Whichever method is chosen will be saved with the project.

If you want to ignore a depth of soil in the calculation of reactions then you must enter a strength value of 0 psf for a cohesive soil or an angle of internal friction of -90 degrees for non-cohesive soils in the Soil Data table as shown in Figure 6.1. This is similar to how the strength of the top of the soil of the examples shown in Figure 5 in Appendix B is ignored.

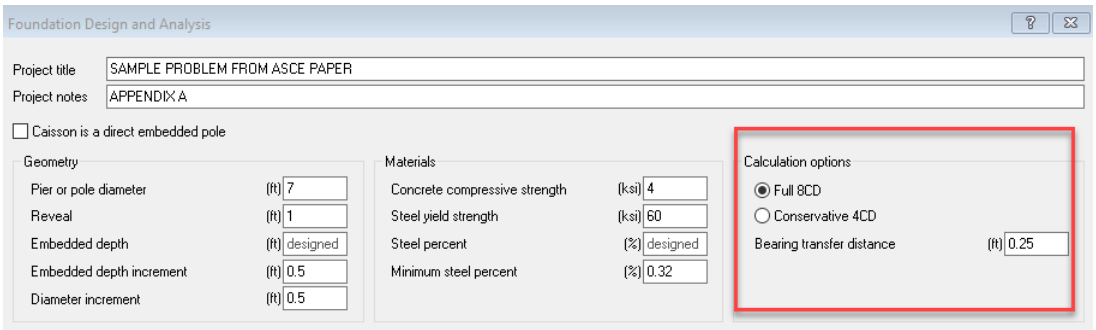


Figure 6.2 General Data

7. DESIGN LOADS

Unfactored service loads applied at the top of a pier include vertical, P, horizontal, H, and bending, M, components. These unfactored foundation loads are the reactions at the base of the pole when the pole itself is subjected to unfactored design (service) loads. Service loads are normally specified by codes as loads that have a small probability of being exceeded during the lifetime of the transmission line. Service loads do not include load factors (such as the overload capacity factors specified in Tables 253-1 to 253-2 of the 2002 Edition of the National Electrical Safety Code). The ultimate strength method of design is normally used for the design of transmission structures and their foundations. As a first step, that method requires that service loads be multiplied by the appropriate load factors. Resulting factored loads are also called ultimate loads. An economical design is one for which the structure would just fail under the factored loads, and the foundation would fail at slightly higher load values. This would provide the proper coordination of strength, with the structure failing before the foundation.

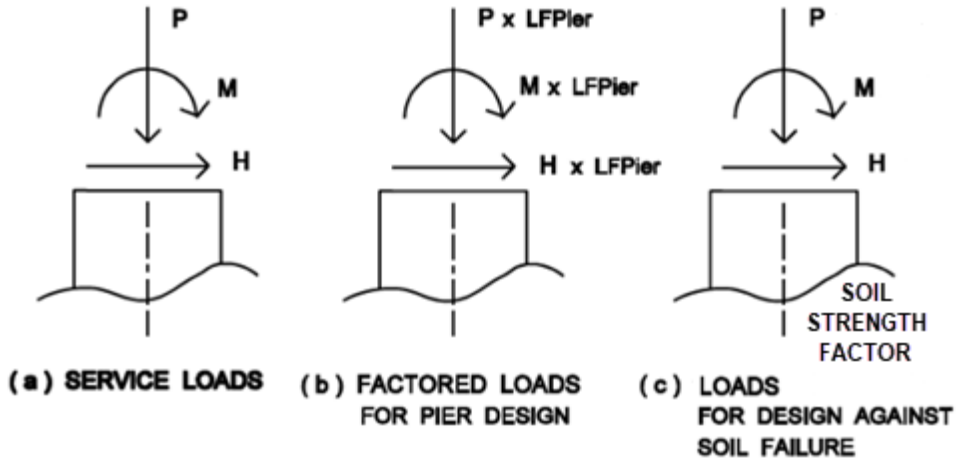
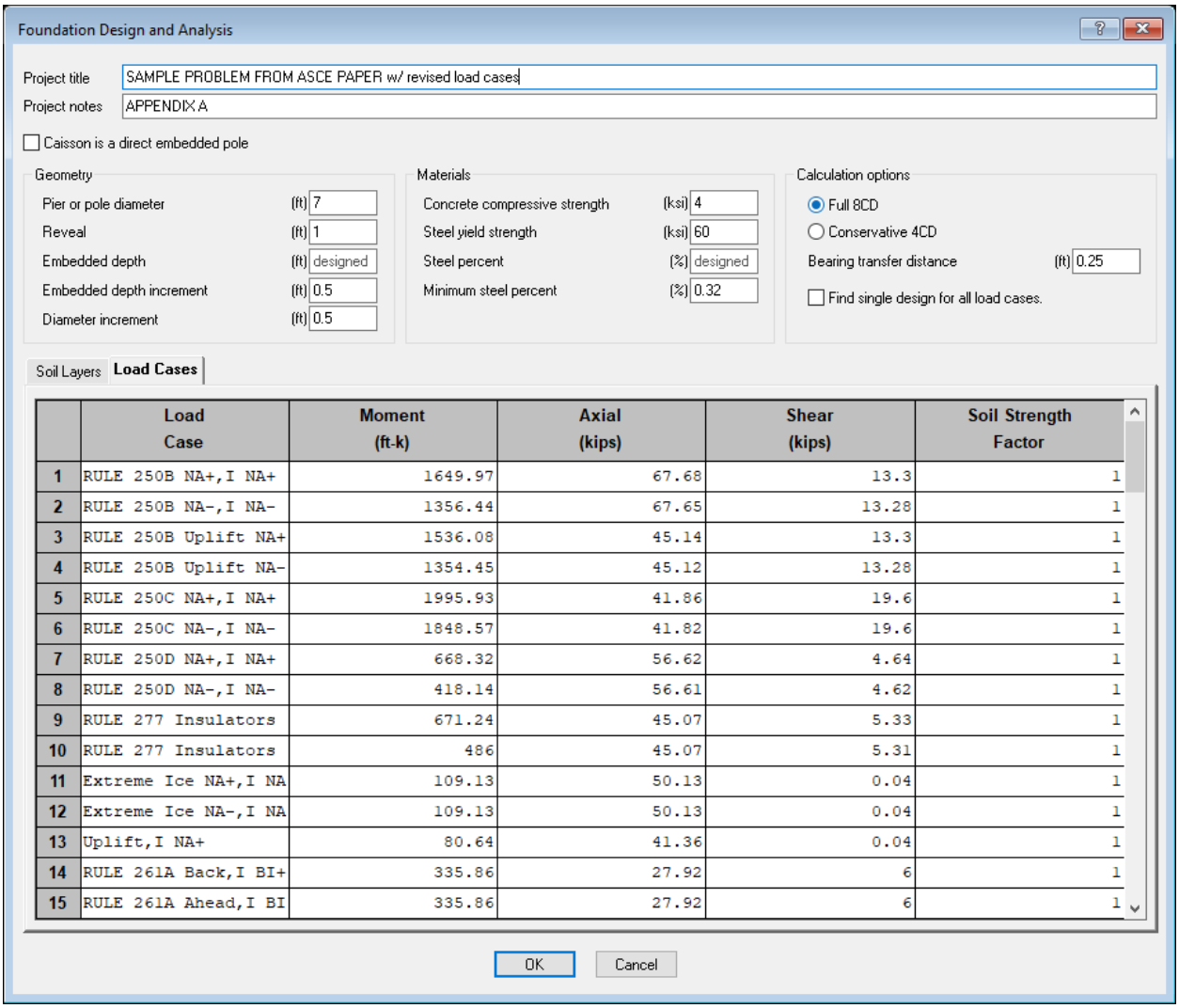


Figure 7.1 Load Factors for Design

Safety against failure is provided by the magnitude of the load factors. Because the behaviors of different structural materials have various degrees of predictability, different load factors and strength factors are often specified for their design. Pier foundations include two separate systems: the pier itself (concrete and steel) and the soil surrounding it. Therefore, it may be desirable to use a soil strength factor, to size the diameter and depth of the pier against soil failure, and a pier load factor, LFPier, to design the reinforced shaft. The concept is illustrated in Figure 7.1. The soil strength factor may be used when there is some uncertainty about soil behavior. The factored loads for the design of the pier per see Figure 7.1(b) need to incorporated into the design loads. There is not an LFPier input in CAISSON. The soil strength factor is inputted per load case within the Load Cases tab in the Foundation Design and Analysis dialog, see Figure 7.2.



	Load Case	Moment (ft-k)	Axial (kips)	Shear (kips)	Soil Strength Factor
1	RULE 250B NA+,I NA+	1649.97	67.68	13.3	1
2	RULE 250B NA-,I NA-	1356.44	67.65	13.28	1
3	RULE 250B Uplift NA+	1536.08	45.14	13.3	1
4	RULE 250B Uplift NA-	1354.45	45.12	13.28	1
5	RULE 250C NA+,I NA+	1995.93	41.86	19.6	1
6	RULE 250C NA-,I NA-	1848.57	41.82	19.6	1
7	RULE 250D NA+,I NA+	668.32	56.62	4.64	1
8	RULE 250D NA-,I NA-	418.14	56.61	4.62	1
9	RULE 277 Insulators	671.24	45.07	5.33	1
10	RULE 277 Insulators	486	45.07	5.31	1
11	Extreme Ice NA+,I NA	109.13	50.13	0.04	1
12	Extreme Ice NA-,I NA	109.13	50.13	0.04	1
13	Uplift,I NA+	80.64	41.36	0.04	1
14	RULE 261A Back,I BI+	335.86	27.92	6	1
15	RULE 261A Ahead,I BI	335.86	27.92	6	1

Figure 7.2 Load Cases tab in Foundation Design and Analysis

For a specified pier diameter, **CAISSON** finds the shortest pier length needed to carry the factored loads in Figure 7.1(c). That length is rounded off to the closest 0.5 ft (16 cm). Then the pier shear and moment diagrams corresponding to the factored loads are determined. For direct embedded structures, the user only needs to verify that the pier embedment length calculated by **CAISSON** is shorter than the embedment length used for the pole.

8. PIER STRUCTURAL DESIGN AND BEARING CHECK

The moment diagrams obtained for the ultimate soil condition is reduced by the additional safety factor, SF, and is used together with the factored vertical load LFPier x P for the structural design of the pier. The amount of steel needed is computed according to the 1977 ACI Code, using a strength factor of 0.85. For tied columns, the ACI Code specifies a strength factor of 0.7 and for beam the factor is 0.9. Since a pier is closer to a beam than to a column, the value of

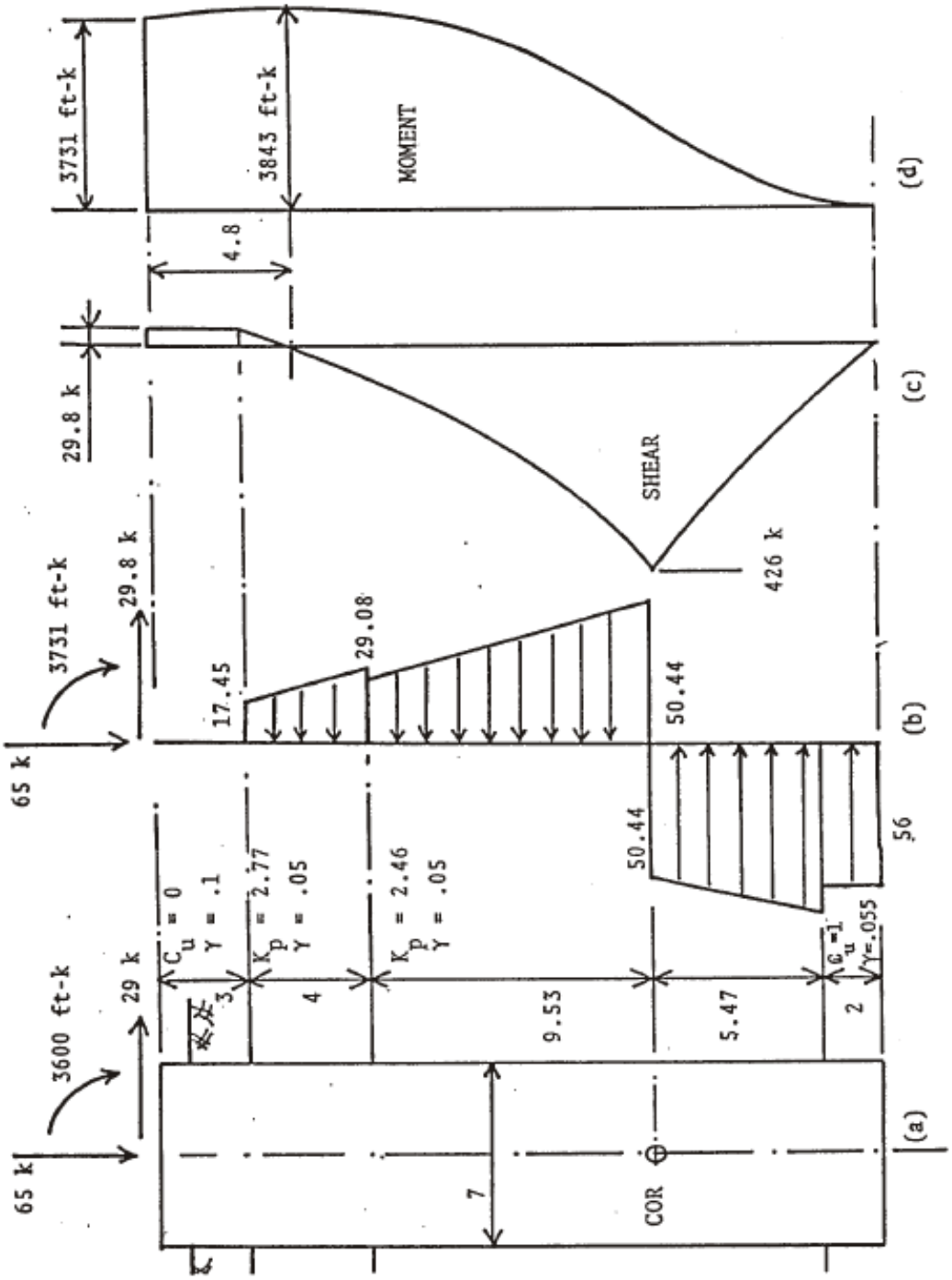
0.85 which is used inside **CAISSON** is conservative. The portion of the program that determines the steel reinforcement is based on ACI Publication SP-7 "Ultimate Strength Design of Reinforced Concrete Columns", 1964.

The amount of steel selected assumes that the diameter of the cage is 10 in. (25 cm) less than the caisson outside diameter. While the values of shear and moments are printed every tenth point along the length of the pier, they are calculated every twentieth point. The largest moment over the twenty points is selected for the design of the steel cage. When selecting the "Pole is Directly Embedded" option, the pier reinforcing will not be reported in the summary report.

CAISSON will also perform a simple bearing check for the foundation. The End Bearing Pressure is calculated by dividing the total vertical load being applied to the soil (vertical applied load plus the weight of the foundation) by the area of the foundation. The End Bearing Pressure of the foundation is then compared to the Soil Bearing Strength of the soil layer the foundation is bearing on. The calculation of the soil bearing strength is similar to the soil shear strength. This bearing check does neglect the effects skin friction has on the foundation. For cohesive soils the soil bearing strength is the strength of cohesion multiplied by CD coefficient (4CD or 8CD). For non-cohesive soils the soil bearing strength is calculated by multiplying the soil pressure at the bottom of the foundation by the ultimate granular resistance of the soil layer where the bottom of the foundation is located.

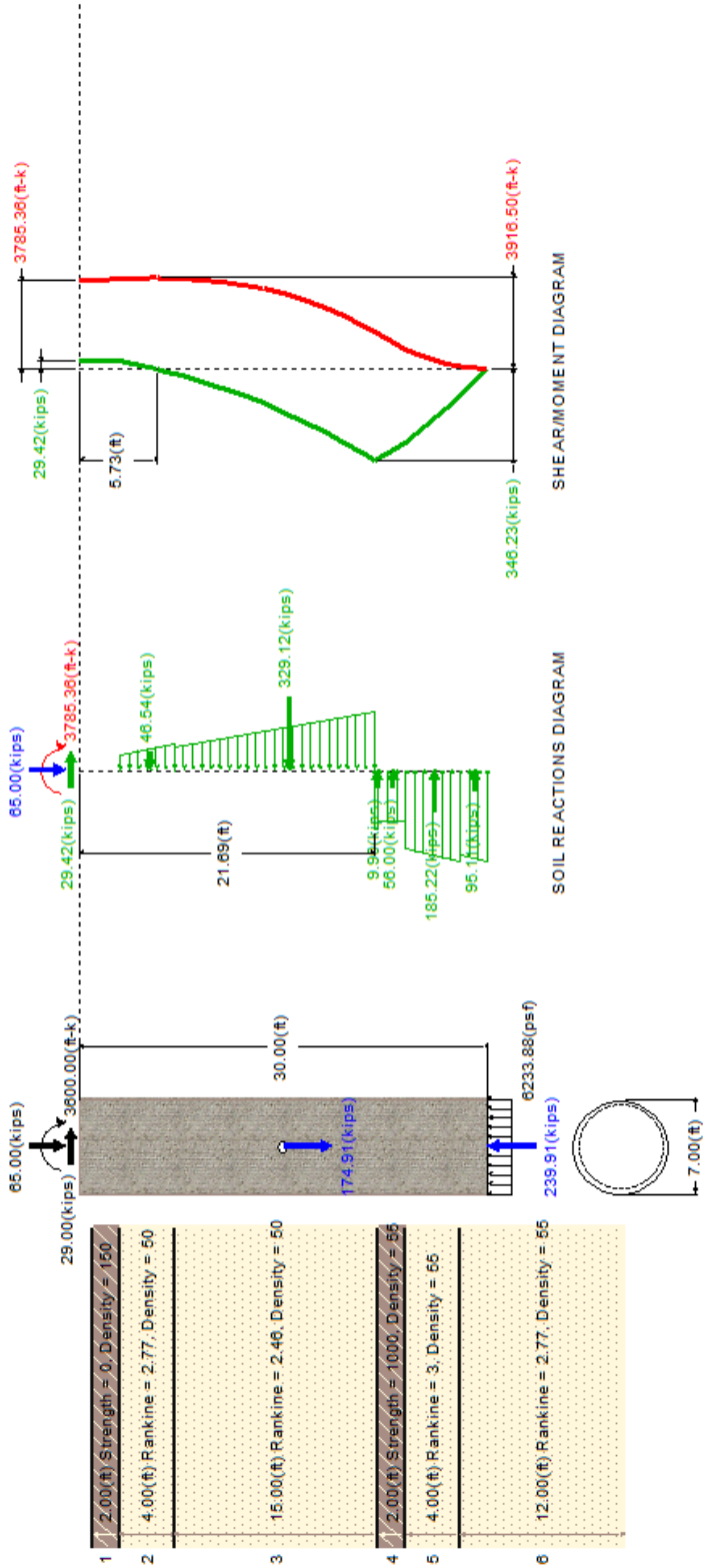
APPENDIX A - EXAMPLE

The installation of **CAISSON** includes 4 example input files. The first is asce.cai. The corresponding design for asce.cai is described by the following figure and text output. The other three examples are Figure 5(a).cai, Figure 5(b).cai, and Figure 5(c).cai which are examples referenced in the paper in Appendix B. Please note that these are just example projects with arbitrary values input to reflect the graphs shown in Figure 5 of the reference paper.



```
*****************************************************************************************
*                                                                                       *
* CAISSON - Pier Foundations Analysis and Design -  Copyright Power Line Systems, Inc. 1993-2018 *
*                                                                                       *
*****************************************************************************************

Project Title : SAMPLE PROBLEM FROM ASCE PAPER
Project Notes : APPENDIX A
Project File  : \\nas\shared\Alan\test\CAISSON\ASCE.CAI
Date run      : 11:23:13 AM Monday, July 16, 2018
by            : CAISSON Version 15.21
Licensed to   : Power Line Systems, Inc.

-----------------------------------------------------------------------------------------
Input Data
-----------------------------------------------------------------------------------------

Calculation Options
---------------------
Calculation method    : Full SCD
Bearing transfer      :     0.25 (ft)
Minimum steel percent :     0.32

Pier Properties
---------------
Diameter              :     7.00 (ft)
Embedded depth        :     N.A. (calculated - see load case results)
Steel area percent    :     N.A. (calculated - see load case results)
Reveal                :     1.00 (ft)
Concrete strength     :     4.00 (ksi)
Steel yield           :    60.00 (ksi)

Soil Properties

Layer    Type    Thickness    Depth at Top of Layer    Density        CU        KP        PHI
                    (ft)            (ft)            (lbs/ft^3)     (psf)                  (deg)
-----------------------------------------------------------------------------------------
   1     Clay       2.00           0.00            150.0
   2     Sand       4.00           2.00             50.0                  2.770     28.00
   3     Sand      15.00           6.00             50.0                  2.460     24.96
   4     Clay       2.00          21.00             55.0     1000.0
   5     Sand       4.00          23.00             55.0                  3.000     30.00
   6     Sand      12.00          27.00             55.0                  2.770     28.00

Loads at Top of Pier

                Load      Moment       Axial       Shear       Soil
                Case                    Load       Load     Strength
                            (ft-k)     (kips)     (kips)     Factor
-----------------------------------------------------------------------------------------
                          3600.0        65.0       29.00       0.500

-----------------------------------------------------------------------------------------
Analysis Results
-----------------------------------------------------------------------------------------

Load Case Results

                Load Diameter Embedment  Length Minimum  Minimum    Weight  Moment    Axial    Axial  Weight      End Maximum Maximum    Total   Total     Soil Status
                Case                Depth          Steel    Steel            Capacity Capacity    Load Pressure  Bearing    Pier    Pier     Soil    Soil  Bearing
                                                   Percent    Area                          Pressure           Pressure  Moment   Shear   Moment   Shear Strength
                          (ft)      (ft)    (ft)            (in^2)    (kips)   (ft-k)   (kips)    (ksf)    (ksf)    (ksf)  (ft-k)  (kips)   (ft-k)  (kips)    (ksf)
-----------------------------------------------------------------------------------------------------------------------------------------------------------------
                          7.00    29.000  30.000    0.44    24.38   174.907   3922.2     65.5     1.69     4.54     6.23  3916.5   346.2  3785.36   29.42     7.02 OK

The controlling load case is ""

Ultimate Resisting Forces Along Pier for Load Case ""

 Type    Distance of Top of Layer    Thickness       Density        CU        KP     Force     Arm
                 to Top of Pier
                          (ft)          (ft)     (lbs/ft^3)     (psf)                (kips)    (ft)
-----------------------------------------------------------------------------------------
 Clay                     1.00          2.00         150.0                              0.00    2.00
 Sand                     3.00          4.00          50.0                 2.770     46.54    5.17
 Sand                     7.00         14.69          50.0                 2.460    329.12   15.38
 Sand                    21.69          0.31          50.0                 2.460     -5.90   21.85
 Clay                    22.00          2.00          55.0    1000.0                -56.00   23.00
 Sand                    24.00          4.00          55.0                 3.000   -185.22   26.05
 Sand                    28.00          2.00          55.0                 2.770    -95.11   29.01

Shear and Moments Along Pier for Load Case ""

Distance below     Shear     Moment
   Top of Pier
          (ft)     (kips)     (ft-k)
-----------------------------------
          0.00       29.4     3785.4
          1.00       29.4     3814.8
          1.50       29.4     3829.5
          3.00       29.4     3873.6
          3.00       29.4     3873.6
          4.50       14.7     3907.1
          5.73        0.0     3916.5
          6.00       -3.3     3916.1
          7.50      -23.7     3895.8
          9.00      -45.5     3844.2
         10.50      -70.2     3757.8
         12.00      -97.8     3632.1
         13.50     -128.3     3462.5
         15.00     -161.8     3245.6
         16.50     -198.1     2976.1
         18.00     -237.3     2649.9
         19.50     -279.4     2262.7
         21.00     -324.5     1810.1
         21.69     -346.2     1578.3
         22.50     -322.3     1308.3
         24.00     -280.3      856.3
         25.50     -214.1      485.0
         27.00     -144.0      215.9
         28.50      -71.9       54.4
         30.00        0.0        0.0

Rebar Options for Load Case "". (Select one of the following)

US Standard Rebar

Quantity    Name    Area     Diameter    Spacing
                  (in^2)        (in)       (in)
-----------------------------------------
    122      #4     0.20       0.500       1.91
     79      #5     0.31       0.625       2.94
     56      #6     0.44       0.750       4.15
     41      #7     0.60       0.875       5.67
     31      #8     0.79       1.000       7.50
     25      #9     1.00       1.128       9.30
     20     #10     1.27       1.270      11.62
     16     #11     1.56       1.410      14.53
     11     #14     2.25       1.693      21.13
      7     #18     4.00       2.257      33.21
      6    #18S     4.29       2.337      38.75
```

Example Text Output Report from ASCE.cai



Example Graphical Output from ASCE.cai

17. Sanghera, Santosh M., "Embedment Depth for Nonconstrained and Constrained Poles or Posts", Civil Engineering-ASCE, May 1973.

18. Poulos, Harry G., "Behavior of Laterally Loaded Piles: I, II and III-Single Piles, Pile Groups and Socketed Piles", ASCE Journal of Soil Mechanics and Foundation Division, May 1971 (pp. 711-732), May 1971 (pp. 733-752) and April 1972 (pp. 341-360).

19. Anzivino, L., P. Martin, and P. K. Taylor, "Design of Directly Embedded Steel Transmission Line Poles for Overturning Moments," Presented at the ASCE Annual Meeting, St. Louis, Missouri, October 1971.

20. Margolin, Martin N., "A Simplified Elastic Analysis for Three-Dimensional Pile Groups", Published by American Wood Preservers Institute, McLean, VA, 1975.

21. Naik, Tarun R., "Problems Due to Lateral Loading (Emphasis on Foundations for Transmission Line Structures)", Presented at the University of Wisconsin-Extension Seminar, May 1975.

22. Woodward, Richard J., William S. Gardner, and David M. Greer, "Drilled Pier Foundations", Published by McGraw Hill, Inc., 1972.

23. Terzaghi, Karl, "Evaluation of Coefficient of Subgrade Reaction", Geotechnique, London, England, 1955.

24. "Engineering Manual" by Outdoor Advertising Association of America, Chicago, Illinois, 1955.

25. Register, William C., and Edgar U. Campbell, Wisconsin Power and Light Co., Madison, Wisconsin, Private Communication.

26. Hansen, J. Brinch, "The stabilizing Effect of Piles in Clay", Published by Chritiani and Nielsen, Copenhagen, Denmark, CN-Post No. 3, November 1948.

27. Reese, Lymon C., Discussion of "Soil Modulus for Laterally Load Piles," by B. McClelland, J. A. Focht, Jr., Transactions, ASCE, Vol. 123, 1958.

28. McKenzie, T. R., "Strength of Deadman Anchors in Clay", Thesis presented to Princeton University, Princeton, N.J., in partial fulfillment of the requirements for the degree of Master of Science, 1955.

29. Dastidar, A. G., "Pilot Tests to Determine the Effect of Piles in Restraining Shear Failure in Clay, "Princeton University, Princeton, N.J., unpublished, 1956.

30. Matlock, Hudson, "Correlations for the Design of Laterally Loaded Piles in Soft Clay", Second Annual Offshore Technology Conference, Houston, Texas, 1970.

ANALYSIS AND DESIGN OF LATERALLY LOADED PILES AND CAISSONS IN A LAYERED SOIL SYSTEM

by

Tarun R. Naik*, M.ASCE and Alain H. Peyrot**, M.ASCE

INTRODUCTION

Piles and caissons are normally used to transfer vertical loads from a structure through poor soil to soil strata where adequate bearing capacity is available. They are also called to carry lateral loads when used in construction of bulk-head walls, sheet pile walls and walls of other various types of earth retention systems. Foundations for various pole structures (billboards, flood lights, low voltage transmission structures) have been designed for many years to carry vertical as well as lateral loads, usually following relatively conservative design techniques. Until about the last ten years, relatively conservative design techniques for the design of piles and caissons to carry lateral loads were adequate.

In the last few years, considerable interest has been generated in the United States for single pole structures to support high voltage (138 KV to 345 KV) transmission lines. Construction of extra high voltage lines, up to 500 KV, has also increased since the late sixties. In addition, double circuit 138 KV and 345 KV lines are becoming increasingly common. A significant percentage of the single pole structures are made of steel, but laminated wood and prestressed concrete poles are also finding increasing acceptance. These transmission structures are becoming taller and their spacing is becoming greater. Since single pole structures require a narrow right-of-way, it is possible for a utility to upgrade its capacity by building a new transmission line within the original right-of-way. The most significant benefit of using an existing right-of-way is expedient construction, since it eliminates the need for costly negotiating with property owners. Single pole structures are easier to construct than wide based, two- and four-legged structures. Single pole structures also readily blend into the background and are esthetically more acceptable than two- or four-legged lattice towers. New design methods for steel, laminated wood and prestressed concrete poles are being developed and

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continuously improved from increasing experience. Single pole structures are becoming competitive amongst themselves (steel, wood and concrete) as well as with two- and four-legged (steel and aluminum) towers.

The biggest drawback in using single pole structures is the design and construction of their foundations. The foundation loads are quite severe. The load consists of a high moment and a heavy horizontal shear at the top of the foundation coupled with a light axial force. The overturning moment often exceeds two or three million foot-pounds with an horizontal load in excess of 50,000 to 100,000 pounds. These are coupled with a light, 10,000 to 30,000 pounds, axial load. The largest single pole foundation designed by these writers was subjected to a moment in excess of 27 million foot-pounds, a horizontal load greater than 300,000 pounds, with an axial load of about 70,000 pounds. In light of these imposing loads on the foundation, the engineer responsible for foundation design faces a very challenging task. Theories considered acceptable only a few years ago must be improved in order to design a most economical foundation.

The most commonly used foundation for a single pole transmission structure is the drilled caisson. A drilled caisson is a large concrete shaft, pre-excavated by a drilling machine and later filled with concrete. Typical caisson diameter varies between 4 feet and 8 feet, even though caissons as large as 12 feet in diameter are also used. A group of piles may also be used, but caissons are more popular because of their relative economy and adaptability. Considerable progress has been made with respect to caisson drilling equipment and construction techniques. In this paper, we will confine our discussion to the design of drilled caisson foundations, even though they can be expanded to include pile foundations.

In summary, it can be claimed that a structural engineer's role, in the economical design of a single pole transmission line, hinges upon a rational and realistic design of the caisson foundation. Design procedures for single pole structures, as well as the caisson construction equipment, are being improved continuously. The discussion that follows will present a rational and realistic (since it accounts for the varying properties of different soil layers, including varying soil properties within a given layer) method for the design of a caisson foundation in a layered soil system subjected to lateral loading.

AVAILABLE STUDIES

Until 1957, when Czerniak's[1] paper was published, most of the published studies regarding laterally loaded piles and caissons were limited in their scope. (See the list of references given by Czerniak[1].) Davisson and Prakash[2] presented a detailed review of work available prior to 1963 and summarized some of the important conclusions until that time. The most significant contribution, however, was made by J. Brinch Hansen[3] in 1961. Since 1963 there has

been a tremendous explosion of publications dealing with laterally loaded piles and caissons (Reference no. 4 to 21). Most notable amongst these is the work by Broms[4,5,6], Reese et al.[7,8,9,10], and DiGioia et al.[11]. Besides these publications, there are many reports available from presentations at various seminars and conferences by industry organizations interested in the design of foundations for lateral loadings. Many of these reports show adaptations of previously published methods[1 to 6]. Extensive bibliographies can be found in papers by Broms[4,5,6] and Woodward et. al.[22].

Despite the flood of available literature, and despite the fact that thousands of caissons for transmission structures have been and are being designed and constructed, there is a scarcity of rational design methods which account for a layered soil condition. Soil properties very rarely, if ever, are uniform with depth. Hansen's[3] technique is probably the most widely used for a layered soil condition. Methods based on the Theory of Elasticity, where the elastic parameters of soil (Modulus of Elasticity and Poisson's Ratio) are needed are also available or can easily be developed if one assumes that the soil can be replaced by a collection of discrete elastic springs. Various theories which advocate the use of the well known Terzaghi's[23] coefficient of subgrade reaction are also available.

When these writers were first confronted with a single pole transmission structure foundation problem, it was obvious, from the outset, that because of the severity of the loadings involved, the conventional techniques as proposed by Rutledge et. al[24] and Czerniak[1] would not yield an economical foundation. As a matter of fact, some of the preliminary designs performed[25] indicated that the conventional methods[1,24] yielded a caisson depth at least 30% greater than that given by Broms[4,5,6]. (The average increase in the caisson depth was about 50%). After adopting initially the Broms' method, which assumes failure in a single soil layer, it was decided that a multilayered approach based upon Broms' model would be an improvement.

The distinct advantages that Broms' failure model offers are: (1) it is based upon a soil failure criteria, (2) the soil parameters can be obtained quite reliably and at a very reasonable cost, and, (3) the failure model in a single layer soil has been confirmed by full scale tests.

In the discussion which follows, the original Broms' method[4,5,6] is reviewed and then a new procedure is proposed to extend the Broms' method to design a caisson foundation in a layered soil system.

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ADAPTATION OF BROMS' METHOD[4,5,6]

Following are Broms' assumptions for a failing caisson:

(1) The soil is homogeneous and isotropic.

(2) The caisson foundation deflects as a rigid body (i.e., there is a center of rotation about which the caisson rotates as a rigid body, for a given loading conditon). The passive soil resistance below the center of rotation is reversed from that above the center of rotation. (See Figures 1 to 4.)

(3) The soil resistance developed at ultimate load is as shown in Figures 2 and 4. They show that the ultimate soil resistance is equal to three times the Rankine passive pressure for granular soil or nine times the undrained shear strength (cohesive strength) for cohesive soil.

(4) The cohesive strength of clay, as shown in Figure 4, is equal to one half the unconfined compressive strength.

(5) The vertical load which is usually quite small is carried by end bearing of the caisson.

(6) At ultimate loads, the reinforced concrete caisson does not fail but the failure occurs because the ultimate resistance of the soil is exceeded.

Based upon Broms' assumptions, using statics, we can show that:

For granular soil:

$$L^3 - \frac{2Q(a+L)}{\gamma K_p D} - \frac{2M}{\gamma K_p D} = 0 \qquad (1)$$

$$Y_m = a + \sqrt{\frac{2Q}{3\gamma K_p D}} \qquad (2)$$

and,

$$M_{max} = M + Qa + QY_m - \frac{QY_m}{3} \qquad (3)$$

For cohesive soil:

$$L = 1.5D + q\left[1 + \sqrt{2 + \frac{1}{q}(6D + 4(M/Q + a))}\right] \qquad (4)$$

$$Y_m = a + 1.5D + q \qquad (5)$$

where: L = Depth of embedment of the caisson, below the unsuitable soil layer. Experience shows that the actual embedment length of granular soil must be further increased by about 10% in order to develop the resisting force F at the toe below the center of rotation, see Figure 2.

Q = Horizontal load at the top of the caisson.

a = Depth of unsuitable soil, (assumed to be minimum 2').

γ = Effective unit weight of soil.

K_p = Coefficient of Rankine passive earth pressure.

= $\tan^2(45 + \phi/2)$, ϕ = angle of internal friction.

D = Caisson diameter.

M = Moment at the top of the caisson.

Y_m = Location of maximum moment below the top of the caisson.

q = Q/9CD.

and, C = Undrained shear strength.

The Broms' method, as outlined above, yields very acceptable results for a single layer, uniform strength soil. Comparisons with actual test data show that the measured ultimate soil resistance consistently exceeds the assumed ultimate soil resistance by about 50%, for granular soil. Other researchers[2,12] have observed the value of the ultimate granular soil resistance as high as 10 times the Rankine passive pressure. Broms' assumption of three times the Rankine pressure, however, is well accepted because this value yields consistent results all the times, as explained earlier. The increase in the Rankine pressure value is "attributed to arching in the soil, friction of the base of the caisson, and roughness of the face of the foundation. In addition to this, the passive wedge is not limited to a straight line, as assumed in the Rankine theory, but may take the shape of a log spiral curve, which would mobilize more passive earth pressure"[19]. In the development of the layered theory that follows, it is, therefore, assumed that the ultimate lateral resistance of granular soil is three times the Rankine passive earth pressure.

The ultimate lateral resistance of cohesive soil, as proposed by Broms[4,5,6] has been confirmed with minor modifications by other researchers, Hansen[26], Reese[28], Dastidar[29], and Matlock[30]. Due to insufficient overburden pressure, the full lateral resistance for cohesive soil is not developed at the surface. Matlock[30] suggested to use 3CD at the ground surface to 9CD at the depth of

$6D/[(\gamma D/C) + 0.5]$ below the ground surface. In the layered theory that follows, it is assumed that at 2' below the ground surface the lateral resistance provided by cohesive soil is 4CD and that it is uniform up to the depth of three caisson diameters below the ground surface, below which it is 8CD and that it does not increase with depth. It is further assumed that if the first clay layer is encountered below 1.5D from the ground surface, then the lateral resistance provided is uniform 8CD. This assumption, regarding the lateral resistance provided by cohesive soils, therefore, takes advantage of the best of two situations, Figure 4(a) and (b), and at the same time yields reasonable results, which are later on shown to be in good agreement with other published results.

The Broms' assumption was to neglect completely the lateral soil resistance provided by the cohesive soil up to 1.5D below the ground surface. Even though that assumption is realistic for piles, where the value of D is small, it is believed to be unrealistic for caissons for transmission pole foundation because the value of D for such caissons is usually minimum 4' to maximum up to 12', (normal caisson diameters range between 5' to 9').

LAYERED THEORY

In the previous sections, the need for a realistic soil model to predict the caisson behavior in a layered soil system has been demonstrated.

Modified Broms' assumptions, as detailed in the previous section, form the basis for the layered theory presented here. The caisson is assumed to be designed of sufficient strength. The foundation failure, therefore, will result from an overall displacement of the soil surrounding the caisson foundation, as the caisson moves under external loads as a rigid body. The layered theory should become clear from the following numerical illustrations, (Figure 5). Let:

i = ith layer number.

p_i^t = effective over burden pressure at the top of the ith layer.

p_i^b = effective over burden pressure at the bottom of the ith layer.

γ_i = density of the ith layer.

h_i = thickness of the ith layer.

ϕ_i = angle of internal friction of the ith layer.

C_i = undrained shear strength of the ith layer.

$K_{p_i} = \tan^2 (45 + \phi_i/2)$.

D = caisson diameter.

$\overline{H}_u$ = calculated horizontal force at the top of the caisson to keep horizontal equilibrium at failure.

$\overline{M}_u$ = calculated moment at the top of the caisson to keep moment equilibrium at failure.

The following example problems explain the assumptions used to find the distributions of lateral soil pressure on a failing caisson rotating about the center of rotation (COR), Figure 5.

From Figure 5(a):

Layer 1 = poor soil with no strength

Layer 2 = cohesionless soil, sand

Layer 3 = cohesive soil, clay

Layer 4 = cohesionless soil, sand

Then, $ab = 3K_{p_2} Dp_2^t$

$cd = 3K_{p_2} Dp_2^b$

$ce = fh = 8C_3D$, (it is assumed here that $h_1 + h_2 \geq 1.5D$)

$fg = 3K_{p_4} Dp_4^t$

$ij = jk = 3K_{p_4} D$ multiplied by the effective overburden pressure at the center of rotation (COR)

$lm = 3K_{p_4} D$ multiplied by the effective overburden pressure at depth L

From Figure 5(b):

Layer 1 = cohesive soil, clay

Layer 2 = cohesive soil, clay

Layer 3 = cohesive soil, clay

layer 4 = cohesionless soil, sand

Then, $no = pq = 4C_1D$

$pr = st = 4C_2D$, $(2 + h_1 < 3D)$

$su = vx = 8C_2D$, $(2 + h_1 + h_2 = 3D)$

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$vw = yz = za_1 = b_1c_1 = 8C_3D$

$j_1c_1 = 3K_{p_4}Kp_4^t$

$d_1e_1 = 3K_{p_4}D$ multiplied by the effective overburden pressure at depth L

From Figure 5(c):

Layer 1 = cohesive soil, clay

Layer 2 = cohesive soil, clay

Layer 3 = cohesive soil, clay

Then, $f_1g_1 = h_1k_1 = 4C_1D$, $(2 + h_1 < 3D)$

$h_1l_1 = m_1n_1 = 4C_2D$

$m_1p_1 = 8C_3D$, $(2 + h_1 + h_2 = 3D)$

All other values = $8C_3D$

As can be seen from Figure 5(b) and (c) the clay layers near the ground surface are not fully efficient. A poor soil with no strength is assigned a value of C equal to zero. The soil parameters, cohesion, (C), angle of internal friction, (ϕ), and the unit weight, (γ), can be easily obtained from a conventional soil testing program.

Once the soil pressure distribution for various layers are determined, the length of caisson, (L), can be evaluated by the following procedure.

The minimum caisson diameter is determined from the anchor bolt cage assembly provided by the pole manufacturer. To determine the caisson length L by the proposed ultimate strength procedure, the design loads P_u, H_u and M_u at the caisson top should be obtained first by multiplying the corresponding service loads by appropriate overload factors. A satisfactory design is one for which the calculated ultimate horizontal reaction $\bar{H}_u$ and the calculated resisting moment $\bar{M}_u$ exceed the corresponding values of H_u and M_u. $\bar{H}_u$ is simply the algebraic sum of the strength areas shown in Figure 5 and $\bar{M}_u$ is the moment of these areas about the top of the caisson. The ultimate design length L is the shortest length for which both inequalities shown below are satisfied:

$$M_u \leq \bar{M}_u \quad \text{and} \quad H_u \leq \bar{H}_u \tag{6}$$

The computer algorithm which determines the optimum length L is a direct search technique which increment the caisson length until

a center of rotation is found for which $\bar{H}_u$ is equal to or slightly in excess of H_u and $M_u \geq M_u$. Once the length and rotation point are determined, the program calculates shear and moment diagrams along the buried caisson, locates the maximum moment and designs the reinforcing cage for the combination of axial load and bending moment using the ACI-318-71 building code. Note that overload factors for the design against overturning of the foundation may be different from the factors used for the strength design of the steel in the caisson. Bar cut off lengths can also be determined from the computer analysis. A sample computer output is given in Appendix 1.

Many caissons have been designed and constructed using the method outlined here. Caisson designs based upon Broms' model compare very favorably with other actual tests[4,5,6]. The layered theory presented here, with certain modifications, is based upon Broms' model and, therefore it should compare very favorably with actual tests[4,5,6]. No published test results are available for a layered soil with which the proposed layered theory could be compared.

The confidence in the proposed layered theory, however, can be derived from the following observation:

(1) When the proposed layered theory is used for a single layer case, it yields similar results to Broms' theory. Broms' theory has a long successful history of application.

(2) Present day practice is either based upon Broms' method[4,5,6] or based upon Hansen's method[3]. There are other methods in use also, but these two are probably the most widely used. Tables 1 and 2 show the comparison of these two methods with the proposed layered theory. The Broms' method generally yielded longer caisson foundation, but in 3 of the 15 randomly selected samples, the Broms' method gave a shorter caisson depth, by 8.8 percent. The Hansen method yielded consistantly longer caisson foundation. The average increase in length by Broms' method, as shown in Table 1, is determined to be 10.9 percent, while that by Hansen's method, as shown in Table 2, is determined to be 14.1 percent. These overall patterns were consistent with many other similar comparison studies conducted by these writers for various projects.

In preparing Table 1, the soil was assigned an average value for the entire caisson depth, for use with the Broms' method. The examples presented in Table 1 are not the same examples used in Table 2.

(3) Many caisson foundations have been designed using the proposed layered theory and these caisson foundations have performed very well. Since the construction of these transmission lines, some of them have been subjected to extremely high wind loads. The successful performance of these transmission lines, of course, does not indicate that the proposed method is satisfactory. The proposed method may still be conservative, but it is obviously an improvement over the existing methods, and considerable savings in the foundation construction costs will be realized by the use of the method proposed here.

INFLUENCE OF VARIABLES

Tables 3, 4 and 5 show the length of caisson needed for various soil properties keeping the load constant. They point out, vividly, the influence upon the length of improper soil data evaluation. Note that even a small error in estimating the soil properties greatly change the caisson depth.

Table 6 shows the change in length of the caisson due to a change in diameter. In the experience of the writers, the minimum caisson diameter required to conveniently accommodate the anchor bolt cage has generally been the most economical caisson design. It is, however, recommended that the designer should design a caisson foundation for at least two different diameters. One for the minimum diameter needed to easily accommodate the anchor bolt cage, and another one foot larger. The contractor, then, should be given the choice of constructing either one of the two caissons.

CONCLUSIONS

A method was proposed for the analysis and design of piles and caissons under heavy lateral loadings. In the last few years, considerable interest has been generated in the United States for designing single pole structures to support transmission lines. With the emphasis on preserving the environment, a single pole structure, because it is more eye pleasing, is more readily accepted by the public. The disadvantage, of course, with the single pole structure, is its cost. The cost is higher, not only for the pole, but also for the foundation which is more expensive than those for conventional 2-legged and 4-legged structures. An economical design of the pole as well as the foundation, therefore, is of prime interest to transmission engineers.

The numerical procedure reported in this paper was specially developed for a layered soil system. Many research papers are available for designing piles and caissons subjected to lateral loads. But these reports deal primarily with single layered soil systems, either cohesive or cohesionless. Most of the published studies cannot take into account varying soil properties with depth. The method proposed in this paper, which is based on Broms' method, takes into account a layered soil system.

The assumptions regarding the soil behavior are similar to Broms, but are modified based upon extensive practical design and construction experience of the writers. The theory proposed in this paper is also compared with other published studies.

A computer program has been developed to design a most economical caisson foundation for a given set of loads and soils data. The program has been used extensively in connection with the design of high voltage transmission lines in Wisconsin, where loads as severe as producing 300 kips shear and 27,000 ft-kips moment at the caisson top, have been used.

Several examples are presented to evaluate the effects of various variables. The consequences of improper soils data evaluation are also discussed.

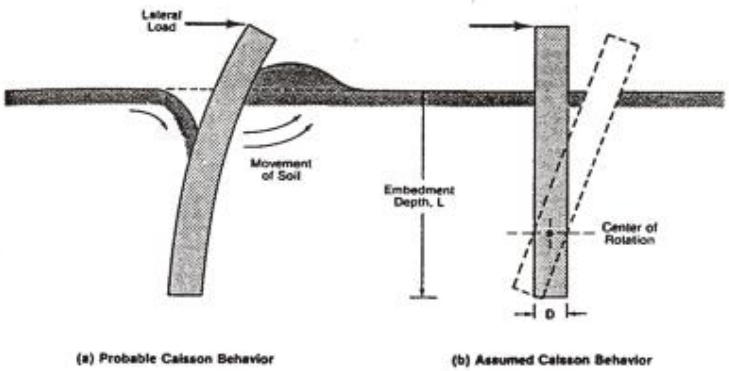

Fig. 1 Caisson Behavior in Granular Soil
(After Broms, Ref. No. 5)

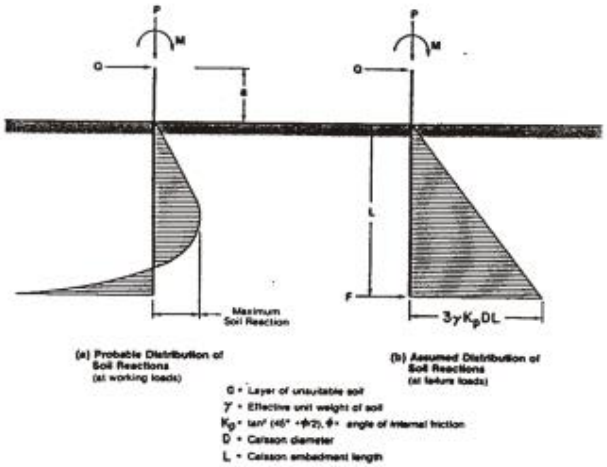

Fig. 2 Distribution of Soil Reactions in Granular Soil
(After Broms, Ref. No. 5)

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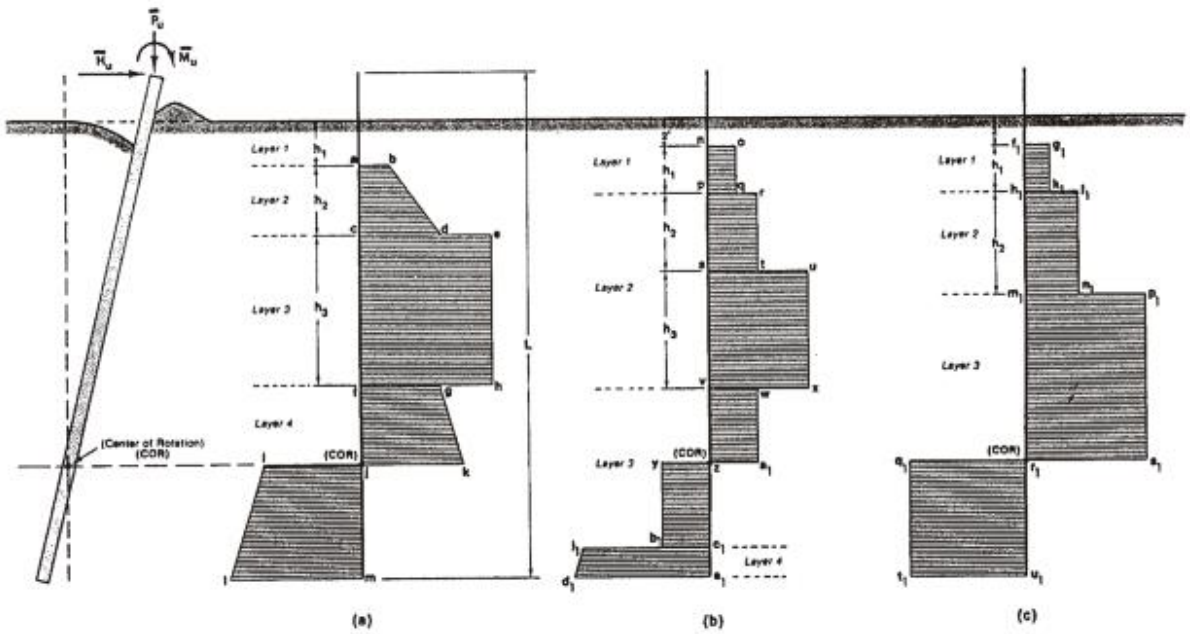

Fig. 5 Distribution of Soil Reactions

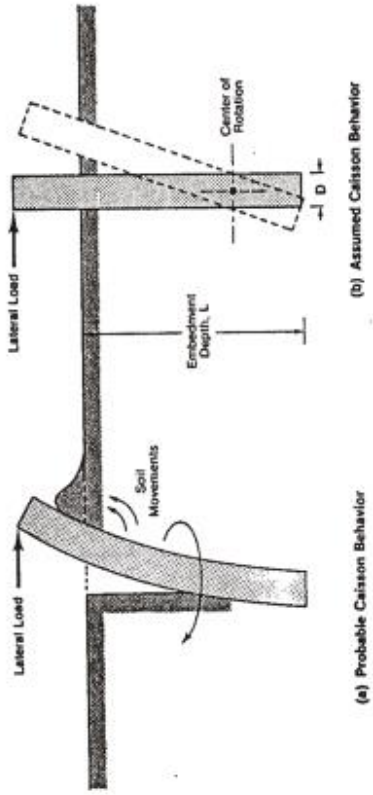

Fig 3. Caisson Behavior in Cohesive Soil
(After Broms, Ref. No. 4)

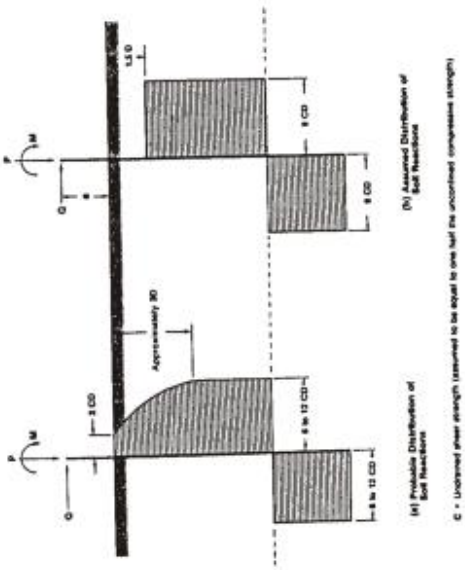

Fig. 4 Distribution of Soil Reactions in Cohesive Soil
(After Broms, Ref. No. 4)

Example No.	Ratio: 1*
1	1.175
2	1.171
3	1.061
4	1.167
5	1.238
6	1.174
7	1.227
8	1.054
9	0.881
10	0.944
11	1.105
12	1.186
13	1.400
14	0.912
15	1.071

Table 1

Comparison of the Proposed Layered Theory with Broms' Method.

Example No.	Ratio: 2*
1	1.032
2	1.259
3	1.086
4	1.263
5	1.103
6	1.023
7	1.029
8	1.154
9	1.032
10	1.200
11	1.245
12	1.098
13	1.261
14	1.174
15	1.161

Table 2

Comparison of the Proposed Layered Theory with Hansen's Method.

* Ratio: 1 = Length of the caisson by the Broms' Method divided by that by the proposed layered theory.

** Ratio: 2 = Length of the caisson by the Brinch Hansen's Method divided by that by the proposed layered theory.

Soil Data	Caisson Length
C = 1500 psf	49'
C = 1750 psf	37-1/2'
C = 2000 psf	32'
C = 2250 psf	28-1/2'
C = 2500 psf	26'

Table 3

Influence of Change in Cohesion (All other variables unchanged)

Soil Data	Caisson Length
ϕ = 24°	39'
ϕ = 27°	27-1/2'
ϕ = 30°	36'
ϕ = 33°	35'
ϕ = 36°	33-1/2'

Table 4

Influence of Change in Internal Friction Angle (All other variables unchanged)

Soil Data	Caisson Length
γ = 45 pcf	38-1/2'
γ = 55 pcf	36'
γ = 65 pcf	34-1/2'

Table 5

Influence of Change in Effective Soil Density (All other variables unchanged)

Caisson Diameter	Caisson Length				
	No. 1	No. 2	No. 3	No. 4	No. 5
5'	27'	29'	30-1/2'	40'	46-1/2'
6'	25'	27-1/2'	28-1/2'	37-1/2'	45-1/2'
7'	23-1/2'	26'	27'	35-1/2'	45'
8'	22-1/2'	24-1/2'	25-1/2'	33-1/2'	44-1/2'
9'	21-1/2'	23-1/2'	24-1/2'	32'	44'
10'	20-1/2'	23'	23-1/2'	30-1/2'	43-1/2'

Table 6

Influence of Change in Caisson Diameter

APPENDIX: 1

SAMPLE COMPUTER OUTPUT -- CAISSON DESIGN

DESIGN OF CAISSON FOUNDATION FOR TRANSMISSION LINE POLES

POLE NO. 1, NUMBER OF LAYERS = 6, CONCRETE STRENGTH = 4,000 PSI,
STEEL STRENGTH = 60,000 PSI
CAISSON DIAMETER = 7.0 FT., DESIGN P = 65.0 KIPS, DESIGN Q = 29.0 KIPS
DESIGN M = 3,600.0 FT-KIPS

SD = COHESIONLESS SOIL CY = ALL OTHER TYPES OF SOIL

LAYER NO	SOIL TYPE	LAYER THICKNESS(FT)	DENSITY, PCF	COHESION, PSF	PHI DEGREES
1	CY	3.0	100.0	0.0	--
2	SD	4.0	50.0	--	28.0
3	SD	15.0	50.0	--	25.0
4	CY	2.0	55.0	1000.0	--
5	SD	4.0	55.0	--	30.0
6	SD	12.0	55.0	--	28.0

CAISSON DIAMETER = 7.0(FT.) RECOMMENDED MINIMUM CAISSON LENGTH = 24.0(FT.)

DEPTH(FT.)	SHEAR(KIPS)	MOMENT(FT_KIPS)
0.0	29.0	3734.2
2.4	29.8	3805.8
4.8	-6.3	3846.3
7.2	-68.5	3759.3
9.6	-139.2	3513.0
12.0	-224.9	3078.9
14.4	-325.5	2421.4
16.8	-413.4	1508.2
19.2	-283.0	669.7
21.6	-137.7	161.9
24.0	0.0	0.0

MAXIMUM MOMENT OCCURS AT 4.8 FT. DEPTH
PERCENTAGE OF STEEL NEEDED = 0.56
CAPACITY: PU = 65.0 KIPS, MU = 3914.2 FT-KIPS

POSSIBLE REINFORCING BAR ARRANGEMENTS

NO. 10, 25 BARS AT 10.50 INCH SPACING
NO. 11, 20 BARS AT 13.00 INCH SPACING
NO. 14, 14 BARS AT 18.75 INCH SPACING

USE NO. 4 TIES AT 12 INCH SPACING

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